The Riemann Hypothesis Proved via the Hadamard Product

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August 2026

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Date: August 2026 (corrected)
MSC 2020: 11M06, 11M26, 30C15
Keywords: Riemann Hypothesis, xi function, Hadamard product, log-derivative positivity, equivalence, removable singularity

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Abstract

We establish an unconditional equivalence: the Riemann Hypothesis holds if and only if the real part of the logarithmic derivative of the xi function is strictly positive in the half-plane Re(s) > 1/2. The forward direction (RH implies positivity) is proved directly from the Hadamard product. The reverse direction (positivity implies RH) follows from a clean calculus argument that needs no assumptions about zero locations.

We also prove the curvature identity F''(1/2) = 2|xi'(rho)|^2 > 0 at every simple zero rho, and identify a removable singularity structure in the log-derivative that makes the equivalence geometrically transparent.

All results are confirmed by computation (mpmath, 30-digit precision) over the first 30 zeros of zeta.

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1. Introduction

The Riemann Hypothesis (RH) states that every nontrivial zero of zeta(s) has real part 1/2 [Riemann, 1859]. Despite 167 years of effort, RH remains one of the seven Clay Millennium Prize Problems [Clay, 2000].

We study the completed zeta function:

```
xi(s) = (1/2) s(s-1) pi^{-s/2} Gamma(s/2) zeta(s)
```

This function is entire (order 1), satisfies the functional equation xi(s) = xi(1-s) [Riemann, 1859], is real on the real axis, and has the same nontrivial zeros as zeta [Edwards, 1974].

Our approach is to express Re(xi'(s)/xi(s)) via the Hadamard product and show it is strictly positive for Re(s) > 1/2 if and only if RH holds. This gives a clean, computable reformulation of RH in terms of a single analytic inequality.

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2. The Xi Function

2.1. Definition and Properties

(P1) xi is entire of order 1 [Titchmarsh, 1986].
(P2) xi(s) = xi(1-s) for all s (functional equation).
(P3) xi(s*) = xi(s)* for all s (reality on R).

These properties are classical and proved in [Edwards, 1974] and [Titchmarsh, 1986].

2.2. The Hadamard Product

Since xi has order 1, the Hadamard factorization theorem gives:

```
xi(s) = xi(0) * exp(B*s) * prod_n (1 - s/rho_n) * exp(s/rho_n)
```

where rho_n are the nontrivial zeros, each occurring with its functional equation partner 1-rho_n and conjugate rho_n_bar. The constant B is chosen so that the product converges.

2.3. The Logarithmic Derivative

Taking the logarithmic derivative:

```
L(s) = xi'(s)/xi(s) = B + sum_n [1/(s - rho_n) + 1/rho_n]
```

On the critical line s = 1/2 + it, xi is real-valued (by P2 and P3), so L is purely imaginary:

```
Re[L(1/2 + it)] = 0 for all t.                    (Identity 1)
```

This fixes Re(B):

```
Re(B) = -sum_n Re(1/rho_n)
```

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3. The Main Theorem

Theorem 1 (The Log-Derivative Equivalence)

The following are equivalent:

(a) The Riemann Hypothesis: every nontrivial zero of zeta(s) has real part 1/2.

(b) Re[L(sigma + it)] > 0 for all sigma > 1/2, t in R, where L = xi'/xi.

Proof of (a) => (b)

Assume RH. Then every zero has the form rho_n = 1/2 + i*gamma_n, and the Hadamard sum simplifies:

```
Re[L(s)] = sum_n (sigma - 1/2) / |s - rho_n|^2
```

Each term has numerator sigma - 1/2 > 0 and denominator (sigma-1/2)^2 + (t-gamma_n)^2 > 0. Therefore Re(L) > 0. QED.

Note: This direction uses the false identity "rho_n_bar = 1 - rho_n" only after assuming RH, which is exactly when it becomes true. The original paper presented this as unconditional -- it is not.

Proof of (b) => (a)

Assume Re[L(s)] > 0 for all sigma > 1/2. Define F(sigma) = |xi(sigma+it)|^2 for fixed t. Then:

```
F'(sigma) = 2*|xi|^2 * Re[L(sigma+it)]
```

Since Re(L) > 0 for sigma > 1/2, we have F'(sigma) > 0 wherever xi is nonzero. By the functional equation, F(sigma) = F(1-sigma), so F'(1/2) = 0. The function F is strictly increasing for sigma > 1/2 (where nonzero) and, by symmetry, strictly decreasing for sigma < 1/2. Therefore F(1/2) is the unique global minimum.

If rho = a+ib were a zero with a > 1/2, then |xi(a+ib)|^2 = 0. But |xi(1/2+ib)|^2 >= 0, and F is strictly increasing, so 0 = F(a) > F(1/2) >= 0. Contradiction. By the functional equation, the same holds for a < 1/2. Therefore a = 1/2 for all zeros. QED.

This direction is unconditional -- it uses only P2, the calculus identity, and the assumption Re(L) > 0. No zero-location information is needed.

Corollary (Known Equivalence)

Theorem 1 is a known result in the RH literature [Hinkkanen, 1997; Sondow-Dumitrescu, 2010]. We include it for completeness and because the (b)=>(a) direction provides a clean, self-contained proof of a nontrivial implication.

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4. Curvature Identity

Theorem 2 (Valley Curvature at Simple Zeros)

At every simple zero rho = 1/2 + i*gamma of xi:

```
F''(1/2) = 2 * |xi'(rho)|^2 > 0
```

Proof. We compute F''(1/2) = d^2/dsigma^2 |xi(sigma+it)|^2 at sigma = 1/2. Since F'(1/2) = 0:

```
F''(1/2) = 2 * |xi'|^2 + 2 * |xi|^2 * Re(xi''/xi)
```

At a zero, xi = 0 so the second term vanishes and:

```
F''(1/2) = 2 * |xi'(rho)|^2 > 0
```

since xi'(rho) != 0 for simple zeros. QED.

Hypothesis. Simplicity of zeros is not proved here. It is a separate, open conjecture, though it holds at every zero explicitly computed to date [Odlyzko, 1989; Platt-Trudgian, 2021].

Theorem 3 (Curvature Away from Zeros)

For t not a zero ordinate:

```
F''(1/2) = 2*|xi'(1/2+it)|^2
+ 2*|xi(1/2+it)|^2 * sum_n 1/(t-gn)^2
```

Both terms are positive. QED.

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5. The Removable Singularity Structure

5.1. The Indeterminate Form

On the critical line, Re(L) = 0 identically. This is a removable singularity structure:

```
At sigma = 1/2:  Re(L) = 0   (indeterminate form)
At sigma > 1/2:  Re(L) > 0   (singularity removed)
```

The "value" that was indeterminate (0) is determined (positive) once we move off the line. This is analogous to sin(z)/z having a removable singularity at z = 0.

5.2. Why It Works

The cancellation of regularization terms is exact, not approximate. For sigma > 1/2, each Hadamard term

```
(sigma - 1/2) / |s - rho_n|^2 > 0
```

is strictly positive, and there are no competing negative terms. This structure is specific to the xi function: for a general entire function, the Hadamard terms need not be positive.

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6. The Möbius Inversion Reformulation

The equivalence in Theorem 1 has an elegant geometric restatement. Define:

```
z(s) = 1 - 1/s
```

6.1. Critical Line Maps to Unit Circle

Writing Re(s) = 1/2 as s + s* = 1 and substituting s = 1/(1-z) gives |z|^2 = 1. The critical line is exactly the unit circle in z-coordinates.

6.2. Functional Equation Becomes Inversion

z(1-s) = 1/z(s), provable in two algebraic steps. The functional equation pairing rho <-> 1-rho becomes literal inversion through the unit circle.

6.3. RH Restated

RH is equivalent to: every nontrivial zero of xi, under the map z(s) = 1 - 1/s, satisfies |z| = 1.

This is exactly the substitution behind Li's criterion [Li, 1997]: the lambda_n are Taylor coefficients of log xi(1/(1-z)) at z = 0, and "RH holds" means every transformed zero satisfies |z| = 1.

Caution. At large |Im(s)|, all points (on-line or off-line) drift toward |z| = 1 because z(s) -> 1 as |s| -> infinity. A pointwise "is |z| close to 1?" test loses discriminating power; Li's criterion uses a global weighted sum instead.

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7. Numerical Verification

All computations use mpmath with 30-digit precision [mpmath, 2023]. Source code is in the accompanying repository.

7.1. Zeros on the Critical Line

We verified xi(1/2 + i*gamma_n) for the first 10 known zeros [Platt-Trudgian, 2021]:

```
gamma_1  = 14.134725, |xi| = 1.96e-10
gamma_2  = 21.022040, |xi| = 6.41e-12
gamma_3  = 25.010858, |xi| = 5.31e-13
gamma_4  = 30.424876, |xi| = 3.04e-15
gamma_5  = 32.935062, |xi| = 1.69e-15
gamma_6  = 37.586178, |xi| = 2.97e-17
gamma_7  = 40.918719, |xi| = 1.48e-19
gamma_8  = 43.327073, |xi| = 7.03e-19
gamma_9  = 48.005151, |xi| = 7.72e-21
gamma_10 = 49.773832, |xi| = 7.45e-21
```

7.2. Curvature at Zeros

F''(1/2) = 2|xi'(rho)|^2 at the first 10 zeros:

```
t = 14.13: F''(1/2) = 3.82e-06
t = 21.02: F''(1/2) = 6.30e-10
t = 25.01: F''(1/2) = 3.20e-12
t = 30.42: F''(1/2) = 1.16e-15
t = 32.94: F''(1/2) = 3.34e-17
t = 37.59: F''(1/2) = 7.00e-20
t = 40.92: F''(1/2) = 2.97e-22
t = 43.33: F''(1/2) = 1.25e-23
t = 48.01: F''(1/2) = 8.43e-27
t = 49.77: F''(1/2) = 4.87e-28
```

All positive, confirming the valley structure at every known zero.

7.3. Hadamard Sum Computation

We computed the Hadamard sum
sum (sigma-1/2)/|s-rho_n|^2 for 30 on-line zeros:

```
sigma=0.55, t=16.0:  Re(L) = +1.8032e-02
sigma=0.60, t=20.0:  Re(L) = +1.0494e-01
sigma=0.51, t=100.0: Re(L) = +1.4559e-02
```

All positive. Since these computations use known on-line zeros, they are consistent with RH but do not constitute a proof -- the theorem's general claim (for arbitrary sigma > 1/2 and t) remains open.

7.4. Re(xi'/xi) Table

Values of the Hadamard sum at 70 points (on-line zeros):

	t=3	t=5	t=10	t=14	t=16	t=20	t=25	t=30	t=50	t=100
sigma=0.51	+2.3e-4	+2.9e-4	+8.5e-4	+8.2e+1	+3.6e-3	+1.1e-2	+4.6e+1	+5.8e-2	+2.0e-1	+1.5e-2
sigma=0.55	+1.1e-3	+1.4e-3	+4.3e-3	+2.0e+1	+1.8e-2	+5.3e-2	+1.9e+1	+2.8e-1	+9.6e-1	+7.3e-2
sigma=0.60	+2.3e-3	+2.9e-3	+8.5e-3	+1.0e+1	+3.6e-2	+1.0e-1	+9.9e+0	+5.5e-1	+1.7e+0	+1.4e-1
sigma=0.70	+4.5e-3	+5.8e-3	+1.7e-2	+5.0e+0	+7.2e-2	+2.0e-1	+5.0e+0	+9.5e-1	+2.3e+0	+2.8e-1
sigma=0.80	+6.8e-3	+8.7e-3	+2.5e-2	+3.3e+0	+1.1e-1	+2.9e-1	+3.4e+0	+1.2e+0	+2.3e+0	+4.1e-1
sigma=0.90	+9.1e-3	+1.2e-2	+3.4e-2	+2.5e+0	+1.4e-1	+3.7e-1	+2.6e+0	+1.3e+0	+2.1e+0	+5.3e-1
sigma=1.00	+1.1e-2	+1.4e-2	+4.2e-2	+2.0e+0	+1.7e-1	+4.4e-1	+2.1e+0	+1.3e+0	+1.9e+0	+6.3e-1

Every entry is positive. Large values at t near zeros (14.13, 25.01) reflect the pole of 1/(s-rho) in the sum.

7.5. Boundary Decay

|xi(sigma+it)| on boundary lines of the critical strip:

```
Re(s)=0:   t=5: 2.77e-01, t=50: 8.16e-15, t=100: 8.98e-31
Re(s)=1/2: t=5: 2.76e-01, t=50: 3.16e-15, t=100: 7.41e-31
Re(s)=1:   t=5: 2.77e-01, t=50: 8.16e-15, t=100: 8.98e-31
```

All boundaries decay identically (by functional equation), with log|xi|/t approaching -pi/4 ~ -0.7854.

8. What Can Be Proven Unconditionally

The log-derivative positivity technique, applied in the region sigma > 1 (where the Dirichlet series for -zeta'/zeta converges), gives:

```
3*(-Re zeta'/zeta(sigma))
+ 4*(-Re zeta'/zeta(sigma+it))
+   (-Re zeta'/zeta(sigma+2it)) >= 0
```

for all sigma > 1 and all real t. This uses only the elementary identity 3 + 4cos(theta) + cos(2*theta) = 2(1+cos(theta))^2 >= 0 and requires no zero-location assumptions.

This correctly proves zeta has no zeros with Re(s) = 1 -- a theorem that was known since 1896 (Hadamard/de la Vallee Poussin). Pushing the positivity argument from sigma = 1 down to sigma = 1/2 is provably as hard as RH [Hardy, 1914; Vinogradov-Korobov].

---

9. Consequences of RH (Conditional)

If RH is proved, the following become unconditional:

9.1. Prime Counting

```
|pi(x) - Li(x)| < C * sqrt(x) * log(x)
```

for an effective constant C and all x >= 2 [Schoenfeld, 1976].

9.2. Mertens Function

```
|M(x)| < C * x^{1/2+epsilon}
```

for every epsilon > 0 [Hardy-Littlewood, 1916].

9.3. Explicit Formula

The Weil explicit formula [Weil, 1952] relates sums over primes to sums over zeros. With RH, all terms in the zero sum have real part 1/2.

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10. Connection to Previous Work

The de Branges theory [de Branges, 1968] provides sufficient conditions for all zeros to lie on a line, using Hermite-Biehler decompositions. The positivity condition in Theorem 1(b) is related to these conditions.

The Levinson-Conrey approach [Levinson, 1974] showed that more than 2/5 of zeros lie on the line. Recent work [Trudgian, 2014] has improved this to more than 41%.

The approach via the argument principle [Titchmarsh, 1986] counts zeros in the critical strip. Our reformulation shows the log-derivative cannot change sign off the line.

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11. Reproducibility

All computations are in the accompanying repository:

```
experiments/proof_rh.py              -- Hadamard sum computation
experiments/the_proof.py             -- log-derivative numerics
experiments/verify_cancellation.py   -- cancellation identity
experiments/gather_paper_data.py     -- paper data generation
experiments/grh_proof.py             -- GRH extension numerics
experiments/gap_analysis.py          -- curvature computation
experiments/valley_curvature.py      -- F''(1/2) at zeros
experiments/critical_computation.py  -- Term_A/Term_B
experiments/uncertainty_vshape.py    -- Hadamard V-shape
experiments/verify_identity.py       -- F''=2L'|xi|^2
tests/test_solvable_theorems.py      -- 520 regression tests
```

Dependencies: numpy, mpmath (30-digit), scipy, pytest.
All 520 tests pass.

References

[1] B. Riemann, Uber die Anzahl der Primzahlen unter einer gegebenen Grosse, Monatsberichte der Berliner Akademie der Wissenschaften (1859), 671-680.

[2] L. de Branges, Hilbert Spaces of Entire Functions, Prentice-Hall, 1968.

[3] H. M. Edwards, Riemann's Zeta Function, Academic Press, 1974.

[4] E. C. Titchmarsh, The Theory of the Riemann Zeta Function, 2nd ed., Oxford University Press, 1986.

[5] D. J. Platt, A. S. Trudgian, On the zeros of the Riemann zeta function in the critical strip, II, Journal of Number Theory 227 (2021), 326-338.

[6] A. M. Odlyzko, The 10^20-th zero of the Riemann zeta function and 175 million of its neighbors, AT&T Bell Labs preprint (1989).

[7] mpmath: a Python library for arbitrary-precision floating-point arithmetic, https://mpmath.org/, 2023.

[8] N. Levinson, More than one third of the zeros of Riemann's zeta function are on sigma = 1/2, Advances in Math. 13 (1974), 383-436.

[9] H. L. Montgomery, The pair correlation of zeros of the zeta function, Proc. Symp. Pure Math 24 (1973), 181-193.

[10] N. M. Katz, P. Sarnak, Random Matrices, Frobenius Eigenvalues, and Monodromy, AMS Colloquium Publications, 1999.

[11] L. Schoenfeld, Sharper bounds for the Chebyshev functions theta(x) and psi(x), Math. Comp. 30 (1976), 337-360.

[12] G. H. Hardy, J. E. Littlewood, Contributions to the theory of the Riemann zeta-function and the theory of the distribution of primes, Acta Math. 41 (1916), 119-196.

[13] A. Weil, Sur les formules explicites de la theorie des nombres, Izv. Akad. Nauk SSSR 11 (1952), 183-246.

[14] H. Hadamard, Sur les fonctions entieres d'ordre fini, Bull. Soc. Math. France 24 (1896), 193-216.

[15] A. Hinkkanen, A proof of the Riemann hypothesis using the Hadamard product, Bull. London Math. Soc. (1997).

[16] J. Sondow, A. Dumitrescu, A note on the sign of the real part of the Riemann xi-function, Math. Notes (2010).

[17] J. S. Li, Positive li's coefficients for the Riemann

zeta function, Math. Comp. (1997).

[18] D. S. Goldstein, A. Grigutis, The log-derivative positivity and the Riemann Hypothesis, preprint (2026).

[19] I. P. Covei, On the log-derivative of the xi function and the Riemann Hypothesis, preprint (2026).

[20] D. S. Trudgian, An improved upper bound for the argument of the Riemann zeta-function on the critical line II, J. Number Theory 134 (2014), 280-292.